QA evidence — v1.0.23 libtinfo cross-distro fix + test-determinism hardening

Run-id: 2026-06-16-libtinfo-crossdistro **Captured:** 2026-06-16 **Hosts:** nezha.local (ALT Linux 6.12 x86_64) · Mistborn (Darwin arm64) **Authority:** §11.4 anti-bluff covenant — every PASS below carries real captured runtime output.

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1. Root cause (FACT, not guess — §11.4.6)

The Linux ELF (built in `ubuntu:22.04`) requires versioned terminfo symbols; the ALT Linux host `libtinfo` has none:

```
# objdump -T tmux  (pre-fix, requires versioned symbols)
(NCURSES6_TINFO_5.0.19991023) setupterm
(NCURSES6_TINFO_5.8.20110226) tiparm
# objdump -T /lib64/libtinfo.so.6  (ALT host — zero NCURSES
version nodes)
<empty>
# ldd tmux  (pre-fix) → warning on EVERY invocation:
tmux/build/bin/tmux: /lib64/libtinfo.so.6: no version information
available (required by tmux)
```

2. Fix proof — nezha (rebuilt binary, post-fix)

```
# ldd tmux/build/bin/tmux   (DT_NEEDED — libtinfo ABSENT, jemalloc
PRESENT)
  libjemalloc.so.2 => /lib/x86_64-linux-gnu/libjemalloc.so.2
  libevent_core-2.1.so.7 => ...
  libm.so.6 / libresolv.so.2 / libc.so.6 / libstdc++.so.6 /
libgcc_s.so.1
# (no libtinfo.so)

# tmux -V  (UNFILTERED stdout+stderr)
tmux 3.6a
# warning lines: 0          ← was 1 per invocation
# ldd | grep -c libtinfo: 0
```

Test 61 (regression guard) — nezha

```
PASS: T1 — binary emits no dynamic-loader version/compat warning
(Linux)
PASS: T2 — tinfo is statically linked (no host libtinfo.so
```

```
dependency)
PASS: T3 — jemalloc dynamic linkage preserved (static-tinfo fix did not regress it)
  Tests: PASS=3  FAIL=0  SKIP=0
```

Test 61 — macOS (Mach-O)

```
PASS: T1 — binary emits no dynamic-loader version/compat warning (Darwin)
SKIP: T2 — libtinfo/ELF symbol-versioning is a Linux-only mechanism (§11.4.81(C))
SKIP: T3 — jemalloc build-time DT_NEEDED check is Linux-ELF-specific
```

verify.sh gate teeth (CM-NO-DYNAMIC-LIBTINFO)

```
real binary libtinfo deps (PASS): 0
system /bin/tmux libtinfo deps (would FAIL): 1
[PASS] CM-NO-DYNAMIC-LIBTINFO (0 dynamic libtinfo deps — tinfo statically linked)
```

3. Full dual-host validation

nezha — final verify.sh (TMX_TEST_DESTRUCTIVE=1)

```
verify EXIT=0
  SUMMARY: PASS=49  FAIL=0  SKIP=11
  GREEN: tmux binary verified — safe to PATH-export.
```

Tests 12 & 14 RAN (not skipped) with real OOM evidence:

```
PASS: T5.1: kernel OOM-kill detected in dmesg (positive evidence)
PASS: T8.1: kernel OOM-kill detected (positive evidence: memory cgroup out of memory)
```

nezha — determinism (§11.4.50), tests 12 & 14 ×5

```
12 run1..5: PASS=3 FAIL=0 SKIP=0   (kernel OOM-kill detected every run)
14 run1..5: PASS=8 FAIL=0 SKIP=0   (scopes B/C survived A's OOM, MainPID unchanged)
```

nezha — meta-test paired-mutation sweep

```
  MUTATIONS CAUGHT (PASS): 52
  MUTATIONS ESCAPED (FAIL): 0
  MUTATIONS SKIPPED:      8
```

Mistborn (macOS) — full suite (isolated TMPDIR, Mach-O binary)

```
EXIT=0
  SUMMARY: PASS=55  FAIL=0  SKIP=5
  SKIPped: 08_oom_score_adj 12_memory_pressure_under_cap
32_ssh_dispatch_remote_nezha
           56_real_mouse_drag_copy 61_no_libtinfo_version_warning
```

4. Test-determinism fixes (RED→GREEN, §11.4.1 / §11.4.43)

Test	RED (pre-fix)	Cause (FACT)	GREEN (post-fix)
27 state-persistence (macOS)	FAIL 3/3 — recall returned stale `-target-path`	`sleep 0.4` raced macOS ~1.5 s cwd-update lag after `send-keys cd`	poll until cd reflected → PASS 3/3
12 memory-pressure (nezha)	FAIL under load — "no OOM-kill in dmesg"	fixed `sleep 2` raced async OOM + `journalctl -k` ingestion lag (dmesg_restrict=1)	poll kernel ring up to ~16 s → PASS 5/5
14 concurrent-OOM (nezha)	FAIL under load — "no OOM-kill detected"	same async/ingestion race (`sleep 10`)	poll up to ~22 s → PASS 5/5

All three are §11.4.1 test-harness timing races (the product worked; the test sampled too early). Assertions unchanged — a genuinely broken feature still FAILs after the full timeout.